Hybrid CMUT and Single Crystal Ultrasonic Transducer Sensor Stack for Muscle Monitoring

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Background, Motivation and Objective

Ultrasound is widely utilized for muscle monitoring in rehabilitation, sports medicine, wearable systems, and neuromuscular diagnostics due to its noninvasive nature, real-time capability, and ability to assess both muscle structure and function. Combining Capacitive Micromachined Ultrasonic Transducers (CMUTs) with Single Crystal Piezoelectrics leverages the high receive sensitivity of CMUTs and the superior transmit efficiency of Single Crystal Piezos. In this setup with separate Transmit (TX) and Receive (RX) elements, coded excitation using long chirp signals further enhances the signal-to-noise ratio (SNR). This phantom study aims to demonstrate that this approach is suitable for muscle monitoring applications.

Statement of Contribution/Methods

A hybrid stack was developed, comprising a single-crystal PMN-PT TX and a CMUT RX chip, each wire-bonded to a separate printed circuit board (PCB) which were attached together, and encapsulated in PDMS (c). Excitation was performed using a Zadoff-Chu chirp (20 µs, 3.5 MHz center frequency, 5 MHz bandwidth), with mismatched filtering, time-gain compensation, and envelope detection for signal processing to obtain A-Scans. Signal generation employed an AMD Zynq RFSoC ZCU216 with amplification via a TI STHV200 chip. The hybrid stack was characterized in degassed deionized water using an aluminum reflector at a 50 mm distance. A custom muscle phantom was constructed from multiple chicken breast strips (3–5 mm thick), which are compressed by a 3D-printed press. For comparison, B-mode images were obtained with a Clarius L15 clinical scanner (a).

Results/Discussion

The hybrid transducer exhibits a center frequency of 3.4 MHz, a −10 dB bandwidth of 5.1 MHz (150 %), and low crosstalk. In muscle phantom experiments, B-mode images reveal visible movement and compression of muscle fibers during pressing (b). The A-Scans from the hybrid stack (d) reflect these changes, with the visible ground reflection (at 5-6 cm) indicating deep penetration and clear extraction of fiber reflections. These results demonstrate that the hybrid approach enables effective muscle monitoring with potentially low energy consumption and minimal channel count (1 TX/RX vs. 196 elements in Clarius), supporting its suitability for wearable and clinical applications.

a) Muscle press with Clarius b) B-Mode (Clarius) c) Hybrid Stack d) A-Scan from Hybrid Stack